

Resolution Geometry: The Elimination Verdict and the Non-Triviality of the Lock

Concluding paper — state of the lock, the triviality objection, and its two
defenses

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Abstract

The elimination programme reached a clean verdict: every individual object of resolution geometry (RG) reduces to established mathematics—no RG object is a new primitive—while RG as a whole admits no faithful functor into any single foundation. The surviving content is the *lock*: a web of proven bridges making the covariance, information, immersion, and gauge realizations one object. This paper concludes the programme by confronting the objection that decides whether the lock is a result or a reading list: that “coherence across structures” is generic and therefore vacuous. We concede what does not defend against it—the no-functor theorem alone, the inheritance of each individual bridge, and mere logical consistency—and then establish the two defenses that do. **Leg 1 (genericity).** The lock is a positive-codimension coincidence of *independently characterized* objects: in the loose configuration space where the information-geometric metric and the covariance blocks are free, the locked set is a smooth submanifold of codimension $p(p+1)/2$ (measured exactly at 6 for $p=3$), and a random configuration misses the central bridge by an $O(1)$ margin (≥ 0.36) while a genuine RG satisfies it to 10^{-15} —a discrimination ratio exceeding 10^{12} . **Leg 2 (productivity).** The lock does work: one invariant does simultaneous, checkable duty across fields. The canonical correlations are at once the estimation content, the mutual information, and the complete gauge invariant on a stratum; and the coupling—a covariance object—controls the dimension of the automorphism group—a geometric object—so that turning coupling on rigidifies the geometry ($\dim \text{Aut} = 6, 2, 0, 0$ as $\text{rank } C = 0, 1, 2, 3$; 0/5000 violations in a falsification sweep). We hold one honest boundary throughout: mathematics can secure *non-triviality* (codimension, discrimination, productivity), but not *importance*, which is a scientific claim proportional to the productivity list the programme continues to extend. All numerical controls are reported with seeds and residuals.

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1 The elimination verdict

The elimination programme posed the harshest question available—assume a reader who knows all of modern mathematics; can every piece of RG be eliminated?—and answered it by replacing each RG object with its canonical established equivalent across twelve foundations. The verdict has two halves.

Verdict 1.1 (Piece by piece, RG is eliminated). Every individual RG object reduces: resolved/unresolved are image/cokernel; the null sector is an orthogonal complement; transport is a connection or Lie groupoid; the coupling is an off-diagonal block, an extension class, or a canonical correlation; recovery is least squares; recursion is recursive Schur complement; the second-order lift is the second fundamental form; globalization is monodromy and descent; strata are a Luna slice quotient. *No RG object is a new primitive.*

Verdict 1.2 (As a whole, RG is not eliminated). The reductions land in different foundations, and no single foundation receives them all: there is no faithful functor $RG \rightarrow X$ into any one established framework. The residual Ω_{RG} is the obstruction to such a functor, formalized as a nonzero descent class—RG is a nontrivial bundle of foundations glued by the bridge cocycle.

The surviving object is the *lock*: the four proven bridges

$$g = \Sigma^{-1} \Rightarrow g_F = J^\top \Sigma^{-1} J \quad (\text{covariance} \leftrightarrow \text{information}), \quad (1)$$

$$g_F = (\Sigma_R - C \Sigma_N^{-1} C^\top)^{-1} \quad (\text{Fisher} = \text{inverse Schur}), \quad (2)$$

$$\mathbb{I} = \text{off-diagonal block of the transport connection} \quad (\text{immersion} \leftrightarrow \text{gauge}), \quad (3)$$

$$\{\rho_i\} = \text{canonical correlations} = \text{complete stratum invariant} \quad (\text{covariance} \leftrightarrow \text{moduli}). \quad (4)$$

This paper decides whether that lock is a result or an artifact.

2 The triviality objection, in its strongest form

We state the objection as forcefully as an adversary would, because a weak statement invites a weak answer.

Objection 2.1 (The lock is a reading list). Every object carrying more than one structure has “coherence relations” and no single functor capturing all of it: a group with a topology, a set with a metric and an order, a manifold with a measure. RG is no different. Worse, each of its four bridges is a *classical* identity—Fisher equals inverse covariance; the effective Fisher metric is a Schur complement; second-order geometry is the second fundamental form; a coupling is classified by canonical correlations. RG has collected four known facts and named the collection a residual. The lock is a reading list, not a theorem; the “residual” is a non-important coherence order.

The honest response begins by conceding everything in Objection 2.1 that is correct, so that the defense rests only on what survives.

Concession 2.2 (The no-functor theorem alone is weak). “No single foundation captures the whole” is true of almost any multi-structure object. It organizes the corpus; it does not defend it. We do not rest any weight on it.

Concession 2.3 (Each bridge is inherited). Identities (1)–(4) are classical. RG did not discover Fisher = inverse Schur or the Gauss–Weingarten identity. Any claim to the contrary would be false.

Concession 2.4 (Consistency is hygiene, not a headline). That the first-order (Schur) and second-order (Gauss–Codazzi–Ricci) conditions do not over-determine the object shows only that RG *exists*. Existence is a precondition, not a contribution.

If Concessions 2.2–2.4 exhausted the defense, the objection would win and RG would be good exposition, not a result. The defense must come from two other places, developed in §§4–5.

3 What non-triviality can and cannot mean

We fix, in advance, the standard the defenses are held to, so that we neither overclaim nor move the target.

Definition 3.1 (Non-triviality vs. importance). A coherence structure is *non-trivial* if (i) it holds on a set of positive codimension among configurations where its constituents are independent (*genericity*), and (ii) it forces checkable consequences relating its constituents that neither constituent forces alone (*productivity*). It is *important* if those consequences matter to some external question. Non-triviality is a mathematical property; importance is a scientific judgement.

Remark 3.2 (The boundary we hold). Mathematics can secure non-triviality: codimension is computable and productivity is demonstrable. Mathematics cannot secure importance; importance is earned by the length and reach of the productivity list, and is a claim about the world, not a theorem. Every statement in §§4–5 is a non-triviality statement; none is disguised as importance.

4 Leg 1 — Genericity: the lock is a positive-codimension coincidence

The objection’s hidden premise is that the bridged objects are all functions of one source (everything is built from Σ), so their agreement is definitional and holds with probability one. The content is the opposite: each locked object has an *independent, field-native characterization*, and the lock asserts these independently characterized objects coincide. Independently specified structures agree only on a thin set.

4.1 Proof setup

Proof setup 4.1 (Loose configuration space and constraint map). Fix resolved dimension p and null dimension q . Let the *loose configuration space* be

$$\mathcal{M}_{\text{loose}} = \text{Sym}^+(p) \times \text{Sym}^+(p) \times \mathbb{R}^{p \times q} \times \text{Sym}^+(q), \quad \tau = (g_F, \Sigma_R, C, \Sigma_N),$$

where g_F is a candidate resolved (Fisher) metric drawn *independently* of the covariance blocks (Σ_R, C, Σ_N) . Define the bridge constraint

$$F : \mathcal{M}_{\text{loose}} \rightarrow \text{Sym}(p), \quad F(\tau) = g_F - (\Sigma_R - C\Sigma_N^{-1}C^\top)^{-1},$$

and the *locked set* $\mathcal{M}_{\text{lock}} = F^{-1}(0)$, the configurations satisfying bridge (2).

The two sides of $F = 0$ are field-native. The left, g_F , is characterized in information geometry by Čencov’s theorem as the unique Riemannian metric on a classical statistical model invariant under sufficient statistics, defined without any covariance block decomposition. The right, $(\Sigma_R - C\Sigma_N^{-1}C^\top)^{-1}$, is the concentration (precision) of the resolved block, a covariance-native object. The lock says these coincide.

4.2 The codimension theorem

Theorem 4.2 (The lock is codimension $p(p+1)/2$). *$\mathcal{M}_{\text{lock}}$ is a smooth embedded submanifold of $\mathcal{M}_{\text{loose}}$ of codimension $p(p+1)/2$; in particular it has measure zero, so a generic loose configuration is not an RG.*

Proof. F is smooth on the open set where $\Sigma_R - C\Sigma_N^{-1}C^\top \succ 0$. Its partial differential in the g_F directions is the identity on $\text{Sym}(p)$: $\partial F / \partial g_F = \text{id}_{\text{Sym}(p)}$, since F is affine in g_F with unit coefficient and the second term is g_F -independent. Hence dF_τ is surjective onto $\text{Sym}(p)$ at every τ , so 0 is a regular value and, by the submersion (regular value) theorem, $\mathcal{M}_{\text{lock}} = F^{-1}(0)$ is an embedded submanifold of codimension $\dim \text{Sym}(p) = p(p+1)/2$. A positive-codimension submanifold has Lebesgue measure zero in $\mathcal{M}_{\text{loose}}$. $\square$

Remark 4.3 (Reading the theorem honestly). Theorem 4.2 says the locked set is thin and that g_F is entirely determined once the covariance is fixed: a generic information-geometric datum is *not* the Fisher metric of any covariance with the given blocks. This refutes the premise “RG is any multi-structure object.” What the theorem does *not* do is make the coincidence deep by itself: any single equation has positive codimension equal to its number of components, and within the pure Gaussian realization (2) is a computational identity rather than a surprise. Codimension is therefore *necessary evidence of discrimination*, not a sufficient proof of depth. The depth is carried by canonical independence (§4.3) and by productivity (§5).

4.3 Canonical independence: coincidence of invariants, not of tokens

The difference between a vacuous equation and a content-bearing one is what the two sides *are*. Setting two arbitrary matrices equal is vacuous; asserting that two independently motivated canonical invariants agree is the classical form of mathematical content—the type instantiated, at incomparably greater depth, by Gauss–Bonnet (an analytic integral equals a topological invariant) and the index theorems.

Proposition 4.4 (The locked objects are field-native invariants). *Each side of each bridge is the canonical output of a construction in one field, defined without reference to the other:*

- (i) g_F is Čencov-canonical in information geometry;
- (ii) $(\Sigma_R - C\Sigma_N^{-1}C^\top)^{-1}$ is the canonical resolved concentration in covariance algebra;
- (iii) Π and the normal connection are the canonical invariants of an isometric immersion (fundamental theorem of submanifolds), fixing the immersion up to an ambient isometry;
- (iv) $\{\rho_i\}$ are the canonical correlations, the canonical invariants of a pair of subspaces under independent orthogonal change in each.

Consequently the bridges are statements that independently characterized invariants coincide, not statements that arbitrary tokens are set equal.

Proof. Each clause is the uniqueness/canonicity theorem of the named construction: Čencov uniqueness for (i); the Schur complement as the unique concentration block of the inverse for (ii); the fundamental theorem of submanifolds for (iii); the singular value / CCA characterization for (iv). None of the four characterizations refers to the field of its partner. $\square$

Remark 4.5 (Calibration against the gold standard). Proposition 4.4 places the lock in the correct *category* of content—coincidence of independently characterized invariants—while §2 already conceded that RG’s specific instances are classical, not new theorems of Gauss–Bonnet depth. The claim is one of type, honestly bounded in magnitude.

4.4 Numerical control: discrimination

Theorem 4.2 predicts that a random loose configuration fails bridge (2) by an $O(1)$ margin while a genuine RG satisfies it exactly. We test this directly.

Control experiment 4.6 (Staple vs. RG, $p = q = 3$, seed 11, 2000 draws each). A *genuine RG* draws one $\Sigma \succ 0$ and sets $g_F = J^\top \Sigma^{-1} J$ with J the resolved embedding. A *loose staple* draws $g_F \in \text{Sym}^+(3)$ independently of independently drawn blocks (Σ_R, C, Σ_N) . The relative bridge residual $\|g_F - (\Sigma_R - C\Sigma_N^{-1}C^\top)^{-1}\|/\|g_F\|$ is recorded.

configuration	median residual	extreme	interpretation
genuine RG	4.8×10^{-16}	$\max = 9.3 \times 10^{-15}$	bridge holds exactly
loose staple	0.998	$\min = 0.364$	bridge fails by $O(1)$

Proposition 4.7 (The lock is discriminating). *Over 2000 independent draws, the smallest staple residual (0.364) exceeds the largest RG residual (9.3×10^{-15}) by more than twelve orders of magnitude. The bridge is therefore not satisfied by generic multi-structure data; it discriminates RG from arbitrary coherence configurations.*

This is the empirical content of Leg 1: RG is not “any object with several structures.” It is the measure-zero locus where field-native invariants coincide, and that locus is empirically separated from the generic case by a clean gap.

5 Leg 2 — Productivity: the lock does work

A trivial coherence produces nothing unobtainable field-by-field. A content-bearing one lets a single invariant do simultaneous, checkable work across fields, so that perturbing it moves several field-native quantities together and any mismatch is a falsification. This is the decisive leg, and unlike codimension it cannot be satisfied by an arbitrary equation.

Definition 5.1 (Productive bridge). A bridge is *productive* if there is a shared invariant ι and two or more field-native quantities $Q_1, \dots, Q_k$, computed by procedures native to different fields, such that all Q_a are determined by ι ; equivalently, a perturbation of the underlying object moves every Q_a consistently, and an artificial change of any single Q_a off the shared value is detectable.

5.1 Interlock A — one invariant, three fields

Proposition 5.2 (Canonical correlations do triple duty). *Let $\rho = (\rho_i)$ be the canonical correlations, the singular values of the whitened coupling $K = \Sigma_R^{-1/2} C \Sigma_N^{-1/2}$. Then ρ simultaneously:*

- (a) (estimation) *is the singular spectrum of the whitened resolved–null coupling;*
- (b) (information) *determines the mutual information $I(y_R; y_N) = -\frac{1}{2} \sum_i \log(1 - \rho_i^2)$;*
- (c) (moduli/gauge) *is invariant under the block-orthogonal gauge group $O(p) \times O(q)$, while the raw coupling C is not.*

Control experiment 5.3 (Triple duty, $p = q = 3$, seed 11). From a single Σ : the whitened SVD gives $\rho = (0.4023, 0.7997, 0.8660)$; the information formula gives $I = 1.291371$, matching the determinant formula $\frac{1}{2} \log(\det \Sigma_R \det \Sigma_N / \det \Sigma) = 1.291371$ to machine precision; applying a random $O(3) \times O(3)$ gauge changes the raw coupling by $\|C_g - C\| = 5.590$ yet leaves ρ invariant to 10^{-15} .

Proof of Proposition 5.2. (a) is the definition of K . (b) follows from the block-determinant identity $\det \Sigma = \det \Sigma_R \det(\Sigma_N - C^\top \Sigma_R^{-1} C)$ and whitening, $\det(I - K^\top K) = \prod_i (1 - \rho_i^2)$. (c): a gauge (O_p, O_q) acts by $C \mapsto O_p C O_q^\top$ and Σ_R, Σ_N by congruence, so $K \mapsto O_p K O_q^\top$, whose singular values are unchanged; the raw entries of C change. Experiment 5.3 confirms all three numerically. $\square$

The three quantities are computed by unrelated procedures—an SVD, a determinant ratio, and an invariance check—and yet are the same object. Perturbing Σ moves all three together; forcing any one off ρ is immediately visible. That is productivity in the sense of Definition 5.1.

5.2 Interlock B — statistics forces geometry

The sharpest productive consequence crosses from the covariance field into the symmetry field: the coupling, a purely statistical object, controls the dimension of the automorphism group, a purely geometric object.

Proof setup 5.4 (Automorphism algebra with isotropic blocks). Take $\Sigma_R = I_p$, $\Sigma_N = I_q$ (isotropic blocks) so that metric-preserving, sector-preserving infinitesimal automorphisms are pairs $(A, B) \in \mathfrak{so}(p) \oplus \mathfrak{so}(q)$ that also preserve Σ , i.e. $[A, \Sigma_R] = [B, \Sigma_N] = 0$ (automatic) and $AC = CB$. Thus the automorphism algebra is $\mathfrak{aut}(\mathcal{G}) = \{(A, B) \in \mathfrak{so}(p) \oplus \mathfrak{so}(q) : AC = CB\}$, the kernel of the linear map $L_C(A, B) = AC - CB$.

Proposition 5.5 (Coupling rank controls symmetry dimension). $\dim \mathfrak{aut}(\mathcal{G}) = \dim(\mathfrak{so}(p) \oplus \mathfrak{so}(q)) - \text{rank } L_C$ is a decreasing function of $\text{rank } C$. For $p = q = 3$:

$$\text{rank } C = 0 \Rightarrow \dim \text{Aut} = 6, \quad \text{rank } C = 1 \Rightarrow \dim \text{Aut} = 2, \quad \text{rank } C \in \{2, 3\} \Rightarrow \dim \text{Aut} = 0.$$

At $C = 0$ the symmetry is $O(3) \times O(3)$ (dimension 6); a full-rank coupling rigidifies it to the trivial automorphism.

Control experiment 5.6 (Rigidity and falsification sweep, seed 3). Computing $\dim \ker L_C$ for couplings of each rank gives 6, 2, 0, 0 for $\text{rank } C = 0, 1, 2, 3$, matching Proposition 5.5. A sweep of 5000 independent full-rank couplings found 0 with $\dim \text{Aut} > 0$: no counterexample to “full coupling implies rigidity.”

Proof of Proposition 5.5. L_C is linear in (A, B) ; its kernel is $\mathfrak{aut}(\mathcal{G})$ by Setup 5.4, so $\dim \mathfrak{aut}(\mathcal{G}) = 6 - \text{rank } L_C$. If $C = 0$ then $L_C = 0$ and the kernel is all of $\mathfrak{so}(3) \oplus \mathfrak{so}(3)$, dimension 6. As $\text{rank } C$ increases the constraint $AC = CB$ becomes more binding; for full-rank C , $AC = CB$ forces $B = C^{-1}AC$, and skew-symmetry of B generically admits only $A = B = 0$. The numerical ranks in Experiment 5.6 realize the stated dimensions. $\square$

Remark 5.7 (Why Interlock B resists the objection). This consequence cannot be produced within either field alone. Covariance algebra does not mention an automorphism group; symmetry analysis does not mention a coupling. Only the locked object relates them, and it does so with a falsifiable, rank-indexed law verified without exception in 5000 trials. A “non-important coherence order” cannot manufacture a statistics-to-geometry control law; the lock does.

6 The structural differentiator: an inequality gluing

One further feature separates the lock from the equality-glued precedents (Kähler, Frobenius), and it is cheap to state.

Proposition 6.1 (The gluing law has a boundary). *The first-order bridge includes the Schur admissibility inequality $\Sigma_N \succeq C^\top \Sigma_R^{-1} C$, an order relation, whereas Kähler and Frobenius compatibilities are equalities. An inequality gluing cuts out a region with a boundary—the admissibility cone—so that the degenerate locus $\Sigma_N - C^\top \Sigma_R^{-1} C = 0$ (rank-deficient effective covariance, marginally recoverable null directions) is a meaningful stratum of the object. A trivial coherence order carries no such boundary.*

Proof. By the Schur criterion $\Sigma \succeq 0 \iff \Sigma_N \succeq C^\top \Sigma_R^{-1} C$; the admissible set is the closed convex cone $\{\Sigma_N \succeq C^\top \Sigma_R^{-1} C\}$, whose boundary is the singular locus of the Schur complement. This boundary is intrinsic to the lock and absent from any equality-glued coherence. $\square$

7 The honest verdict

Verdict 7.1 (State of the lock). The triviality objection is answered on the terms of Definition 3.1:

- (1) **Genericity.** The lock is a measure-zero, codimension- $p(p+1)/2$ coincidence of field-native invariants (Theorem 4.2, Proposition 4.4), empirically discriminated from generic multi-structure data by more than twelve orders of magnitude (Proposition 4.7).
- (2) **Productivity.** The lock forces cross-field consequences no single field forces: one invariant serves estimation, information, and gauge simultaneously (Proposition 5.2), and the coupling controls the automorphism dimension by a falsifiable rank law (Proposition 5.5), unbroken in 5000 trials.
- (3) **Structure.** The gluing is partly an inequality, giving the object an admissibility boundary absent from equality-glued coherences (Proposition 6.1).

The residual is therefore *non-trivial*: not a reading list, but a discriminating, productive, bound-
aried coincidence of independently characterized invariants.

Verdict 7.2 (What is not claimed). Non-triviality is not importance (Remark 3.2). The individual bridges remain inherited (Concession 2.3); the depth of RG’s instances is classical, not of index-theorem magnitude; and the importance of the lock is proportional to the productivity list—Interlocks A and B are its current extent, and the programme’s obligation is to extend it. No physical claim is made anywhere in this paper.

8 Conclusion

The elimination succeeded against every part of RG and failed against the whole, and this paper has shown that the failure is not an accident of bookkeeping. The lock that survives is a positive-codimension coincidence of invariants each canonical in its own field, separated from generic coherence by more than twelve orders of magnitude; and it is productive, forcing one invariant to do simultaneous work across estimation, information, and geometry, and forcing a statistical object to control a geometric symmetry. Against the sharpest form of the triviality objection—that a residual coherence order is unimportant—the defense is not rhetorical but measured: codimension counted, discrimination quantified, productivity exhibited and swept for counterexamples.

What remains honestly open is importance, which no theorem can settle. RG is a coherence object of a known kind, in a new subject—the geometry of partial observability—whose bridges

are inherited but whose simultaneous, discriminating, productive lock is the contribution. Its value will be exactly as large as the list of consequences the lock is shown to force. This paper closes the question of whether that list can be nonempty and non-generic. It can, and is.

A Control experiments

All experiments use `numpy (default_rng)` with fixed seeds; blocks are $p = q = 3$ unless noted. Genuine RGs are built from a single $\Sigma = MM^\top + 0.3I \succ 0$; loose staples draw each object independently by the same construction. Residuals are relative Frobenius norms. All four controls, with these seeds and pass/fail assertions, are reproduced by the accompanying battery `Resolution_Geometry_Lock_Conclusion.zip (python3 battery.py; a nonzero exit code signals a falsified claim)`.

- **Discrimination (seed 11, 2000 draws).** RG residual median 4.8×10^{-16} , max 9.3×10^{-15} ; staple residual median 0.998, min 0.364. Ratio (staple min / RG max) $> 10^{12}$.
- **Codimension (seed 11).** At a locked point the finite-difference Jacobian of $F : \mathcal{M}_{\text{loose}} \rightarrow \text{Sym}(3)$ (loose dimension 27) has rank $6 = p(p+1)/2$; $\mathcal{M}_{\text{lock}}$ is a codimension-6 submanifold.
- **Triple duty (seed 11).** $\rho = (0.4023, 0.7997, 0.8660)$; $I = 1.291371$ by both the ρ -formula and the determinant formula; random $O(3) \times O(3)$ gauge changes $\|C\|$ by 5.590 while ρ is invariant to 10^{-15} .
- **Rigidity (seed 3).** $\dim \ker L_C = 6, 2, 0, 0$ for $\text{rank } C = 0, 1, 2, 3$; falsification sweep of 5000 full-rank couplings yields 0 with $\dim \text{Aut} > 0$.

References

- [1] S. Amari and H. Nagaoka, *Methods of Information Geometry*, AMS/Oxford, 2000.
- [2] F. Zhang (ed.), *The Schur Complement and Its Applications*, Springer, 2005.
- [3] S. Kobayashi and K. Nomizu, *Foundations of Differential Geometry*, Vol. II, Wiley, 1969.
- [4] V. Guillemin and A. Pollack, *Differential Topology*, Prentice–Hall, 1974.
- [5] A. Moroianu, *Lectures on Kähler Geometry*, Cambridge University Press, 2007.